



Introduction to AI – Session I

Nevenka Dimitrova, PhD, MSKCC, USA



PHILIPS



Stanford

MEDICINE



Cima
Universidad
de Navarra



復旦大學 智能医学研究院
Intelligent Medicine Institute, Fudan University

112
1004

Leibniz
Universität
Hannover

AI Before 2000 – Mostly Classical AI and Expert Systems

1. 1950 - Turing Test by Alan Turing

- Proposed by Alan Turing in his paper "Computing Machinery and Intelligence"
- A test to determine if a machine can exhibit intelligent behavior indistinguishable from a human

2. 1956 - Dartmouth Conference

- Organized by John McCarthy, Marvin Minsky, Nathaniel Rochester, and Claude Shannon
- Coined the term "Artificial Intelligence"
- Marked the birth of AI as an academic discipline

3. 1966 - ELIZA

- Created by Joseph Weizenbaum at MIT
- The first chatbot, simulating a conversation with a psychotherapist
- Demonstrated the potential for natural language processing

4. 1997 - IBM's Deep Blue Defeats Garry Kasparov

- IBM's chess computer defeated the reigning world chess champion
- Showcased the capabilities of AI in strategic game playing



After 2000 – Text Analysis, Advanced Neural Networks, Generative AI

5. 2011 - IBM's Watson Wins Jeopardy!

- IBM's Watson competed on the quiz show Jeopardy! against human champions
- Demonstrated advanced natural language processing and information retrieval

6. 2016 - Google's AlphaGo Defeats Lee Sedol

- Google's AlphaGo, developed by DeepMind, defeated Go champion Lee Sedol
- Highlighted the advancements in machine learning and neural networks

7. 2018 - Google DeepMind's first AlphaFold model

- Coming first in an international protein structure prediction competition.

8. 2020 - OpenAI's GPT-3 Released

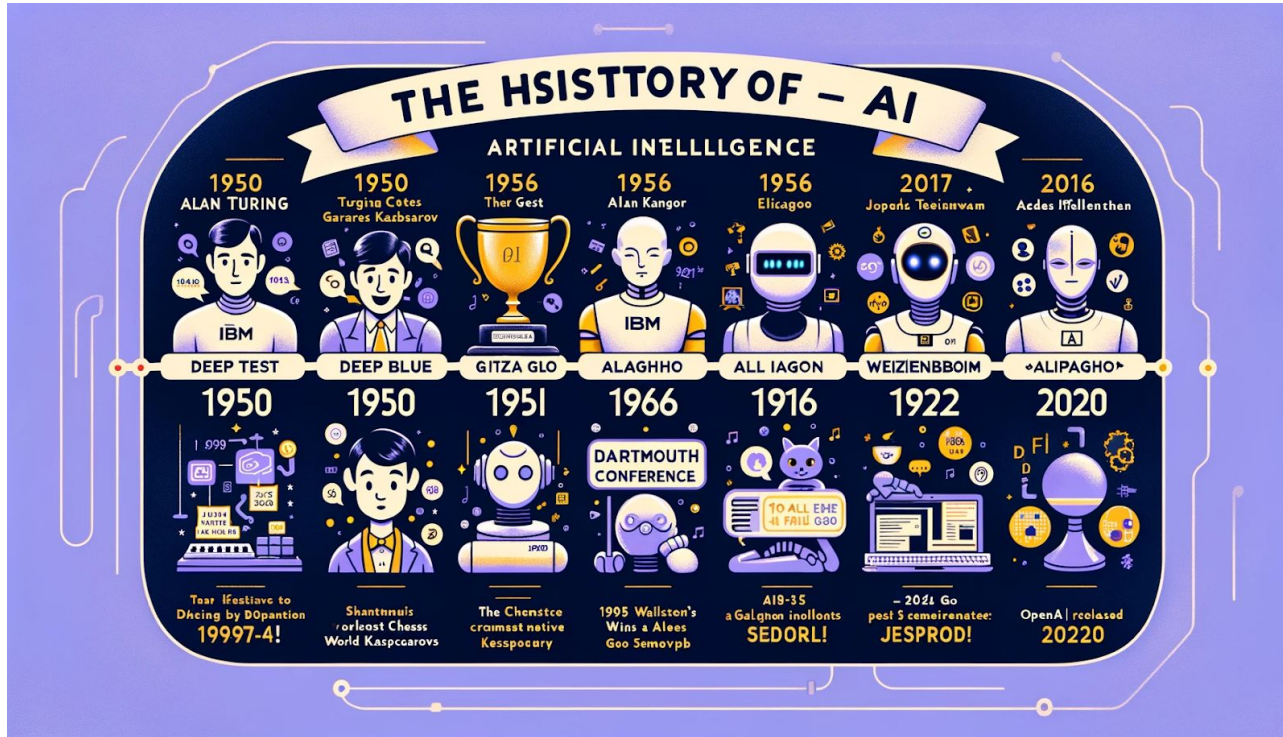
- OpenAI released GPT-3, a state-of-the-art language model
- Showcased the power of deep learning for generating human-like text

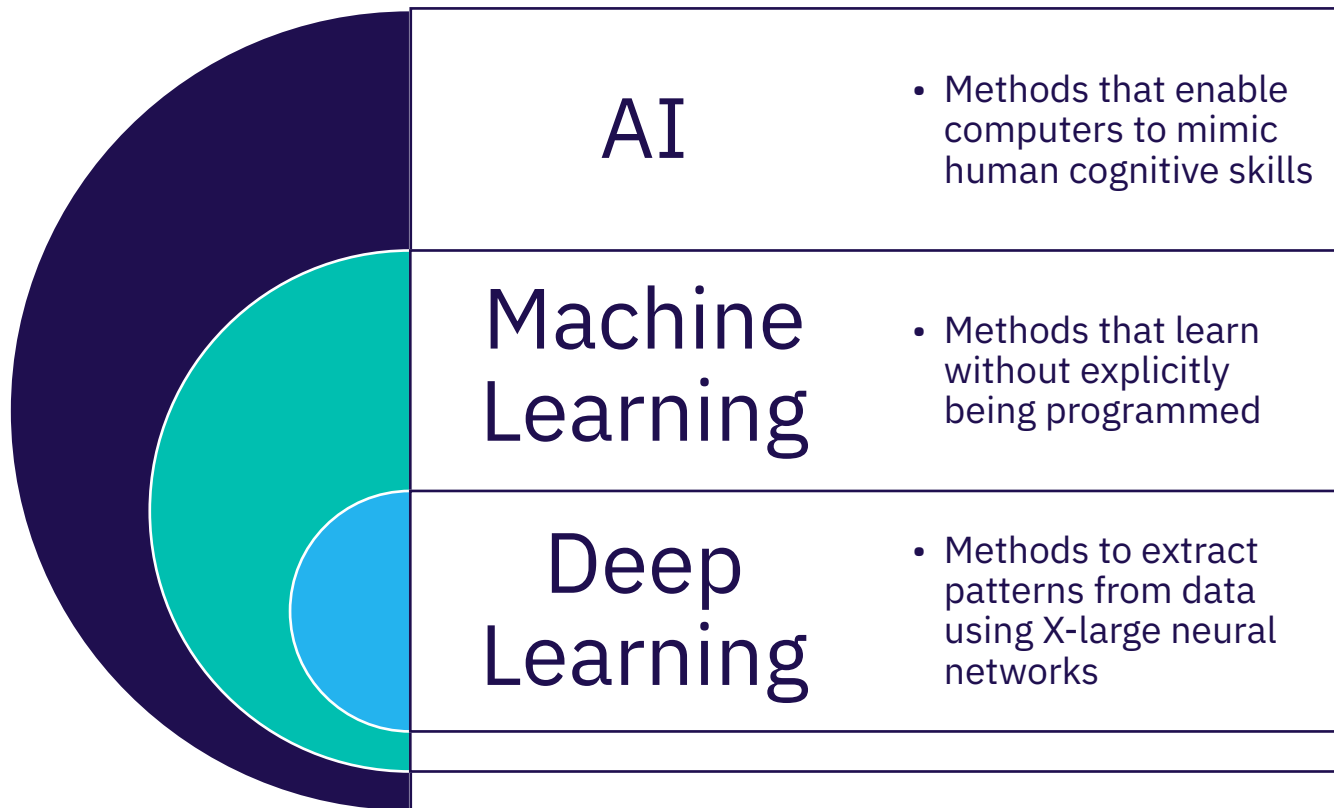
History of AI according to DALLE [edited]

- Left Side: Alan Turing had foundational contributions to AI with the Turing Machine. Early mainframe computers and symbols of the Dartmouth Conference represent the initial stages of AI development.
- Middle : IBM Deep Blue, a robot arm representing industrial robotics, and neural networks illustrate the progress in AI technology over time.
- The right side: humanoid robot, Google DeepMind's AlphaGo, and self-driving cars.



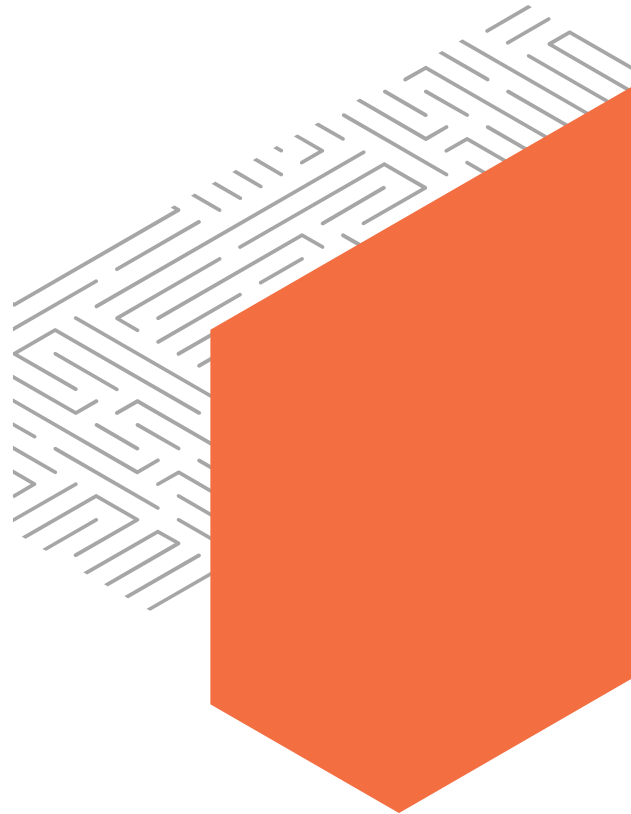
Try not to Laugh! [Created by ChatGPT 4o]





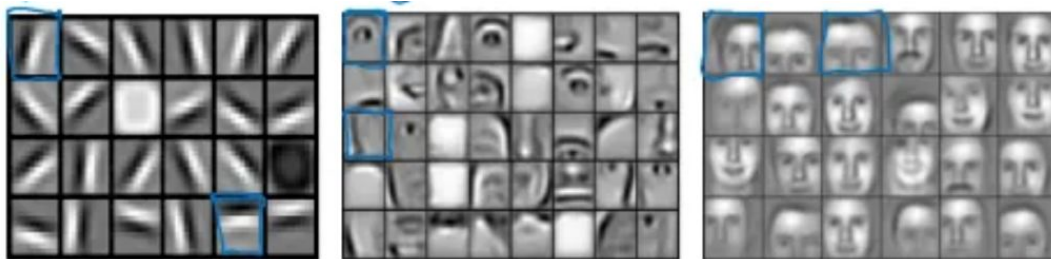
Machine Learning Methods

- Deep learning
- Convolutional Neural Networks • Recurrent Neural networks • Transformers / Large Language models • Autoencoders • Diffusion models...
- Classical ML
- Linear regression • Logistic regression • Perceptron • Naive bayes classifiers • Support vector machines • Convolutional kernels • Decision trees • Hierarchical Clustering • K-nn clustering • Principal Component Analysis • Non-negative matrix factorization...

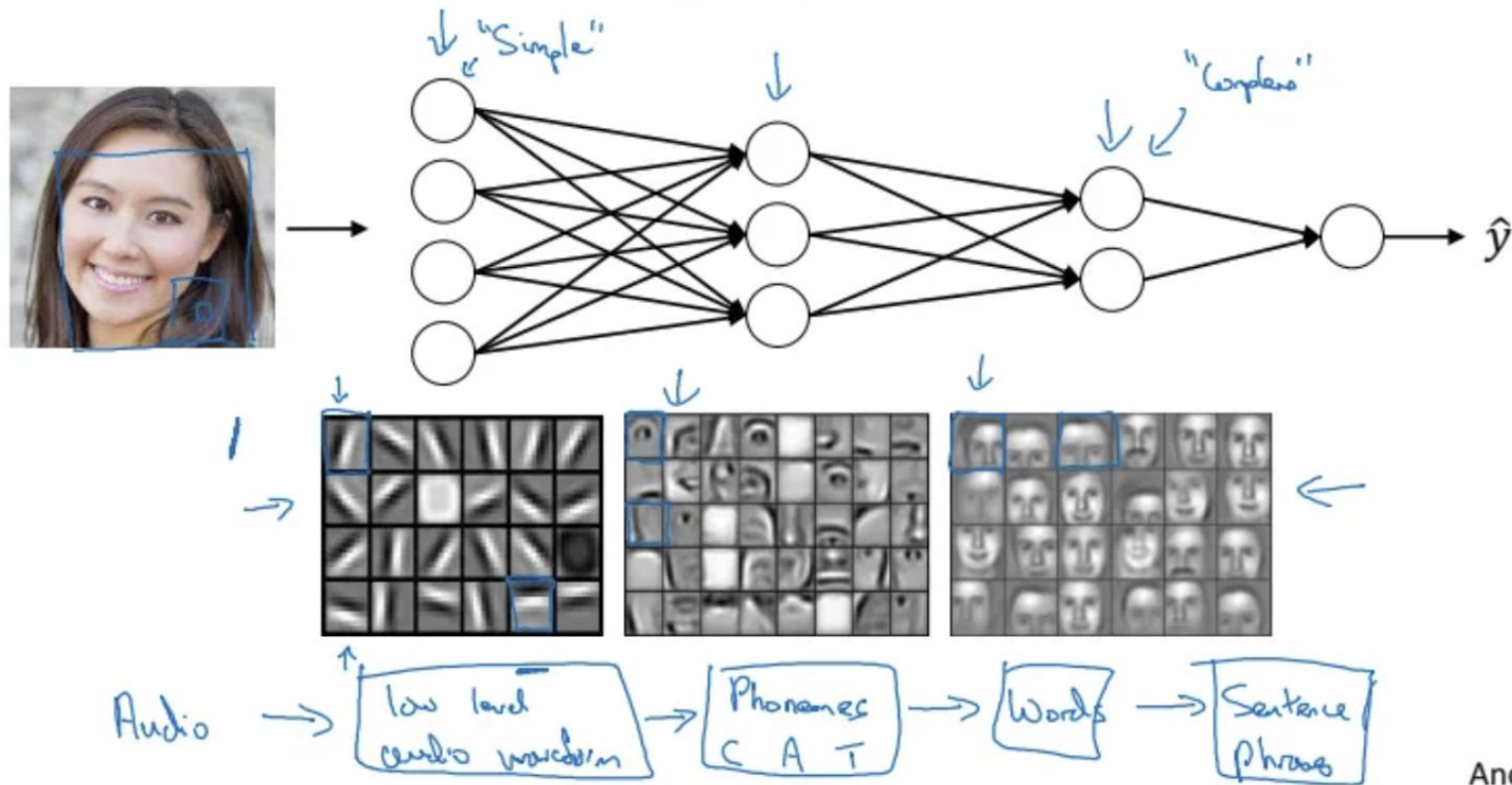


Representations

- Let's represent the statement: "Joe is a parent of Sally"
- Parent (Joe, Sally)
- Let's represent a whole picture:

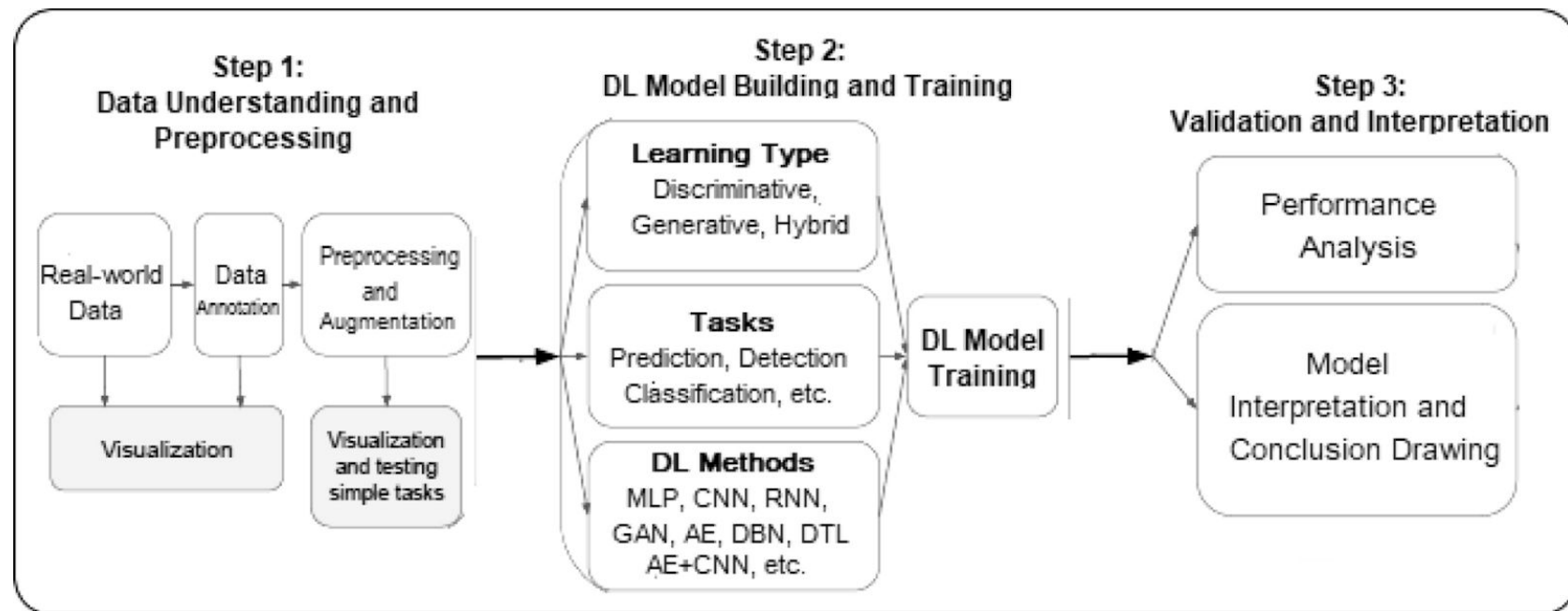


Intuition about deep representation



Andrew Ng

Representations



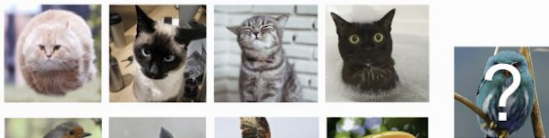
<https://link.springer.com/article/10.1007/s42979-021-00815-1/figures/4>

80:20 rule for supervised training

Training

Train dataset #1: "cat-bird"

cats

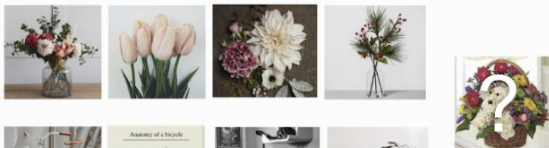


birds



Train dataset #2: "flower-bike"

flowers



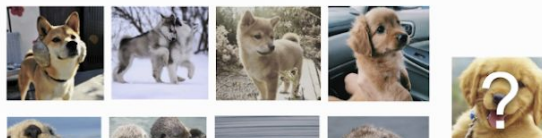
bikes



Testing

Test dataset: "dog-otter"

dogs



otters

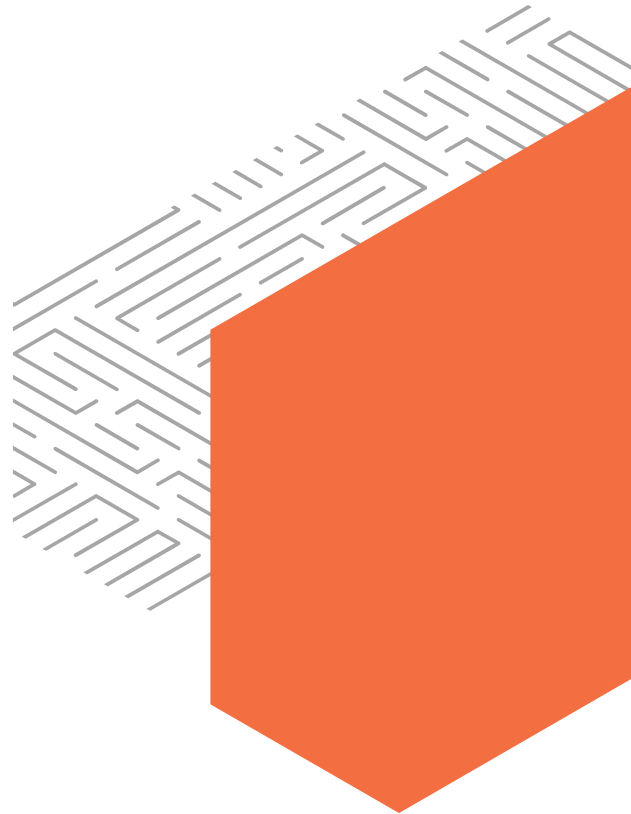


We can embark on this, but!!
We need to cover some basics on neural networks, Python and Pytorch first!!

From single perceptrons to GPTs

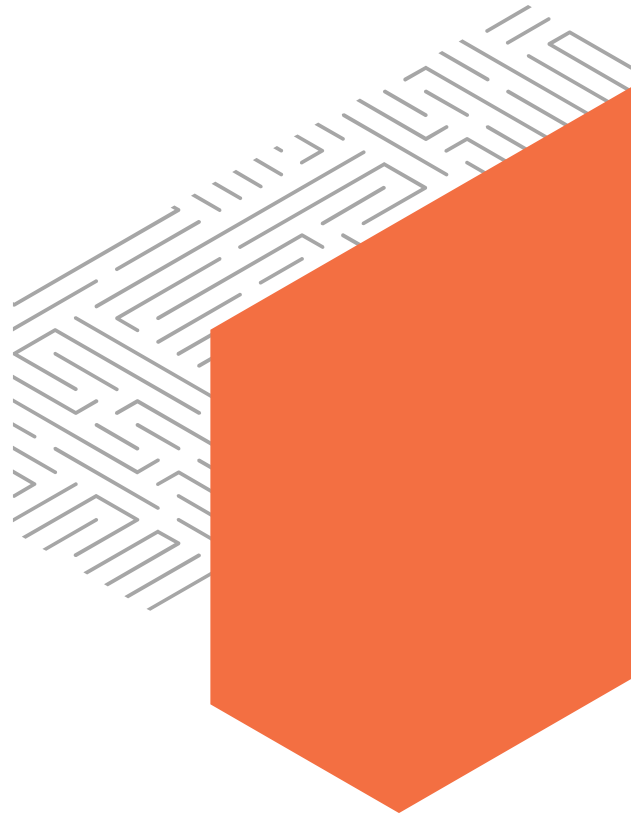
“As an AI language model, I don't have neurons like a biological brain. Instead, I function through a vast network of artificial neurons or parameters in a neural network. Specifically, GPT-4, which powers me, has around 175 billion parameters. These parameters are part of the complex architecture that enables me to understand and generate human-like text based on the input I receive. These parameters are analogous to the weights in a neural network, and they determine how the model processes and generates text.”

– ChatGPT 4o

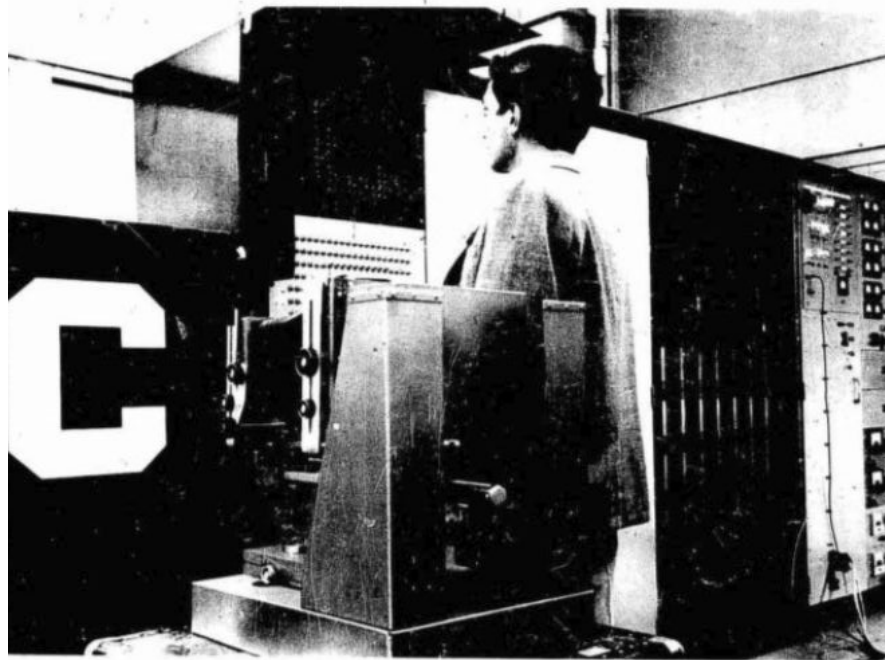


From single perceptrons to GPTs

- The perceptron was invented in 1943 by Warren McCulloch and Walter Pitts. [The first hardware implementation was Mark I Perceptron machine built in 1957 at the Cornell Aeronautical Laboratory by Frank Rosenblatt, funded by the Information Systems Branch of the United States Office of Naval Research and the Rome Air Development Center.]



Mark I



THE MARK I PERCEPTRON

PRINCIPLES OF NEURODYNAMICS

PERCEPTRONS AND THE THEORY OF BRAIN MECHANISMS

BY: Frank Rosenblatt
Frank Rosenblatt
Director, Cognitive Systems Research Program
Cornell University, Ithaca, N. Y.

15 MARCH 1961

The work reported in this volume has been carried out under Contract Nonr-2381(00) (Project PARA) at C.A.L. and Contract Nonr-401(40) at Cornell University. Reproduction in whole or in part is permitted for any purpose of the United States Government.

Mathematical Definition of a Perceptron

The perceptron is an algorithm for learning a binary classifier using a threshold function: a function that maps its input x (a real-valued vector) to an output value $y=f(x)$ which is a single binary value:

Inputs Weights Sum Non-linearity Output

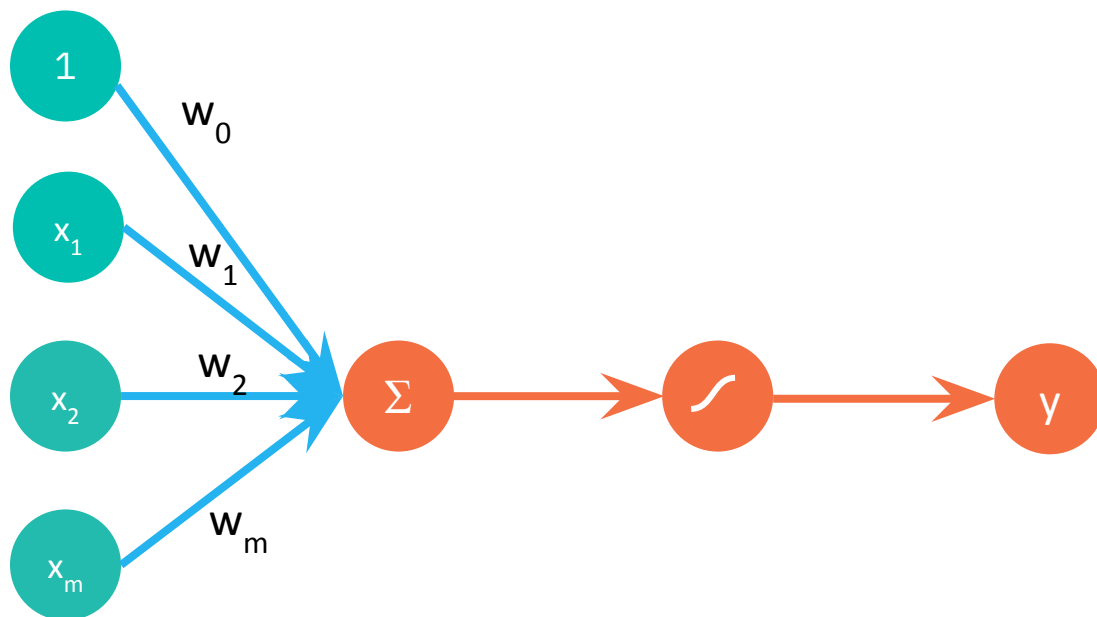
$$y = g(w_0 + \sum_{i=1}^m w_i x_i)$$

$$y = g(w_0 + w_1 x_1)$$

$$y = g(w_0 + w_1 x_1 + w_2 x_2)$$

$$y = g(w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3)$$

$$y = g(w_0 + w_1 x_1 + w_2 x_2 + \dots + w_m x_m)$$



Vector Equation of a Perceptron

$$y = g(w_0 + \sum_{i=1}^m w_i x_i)$$

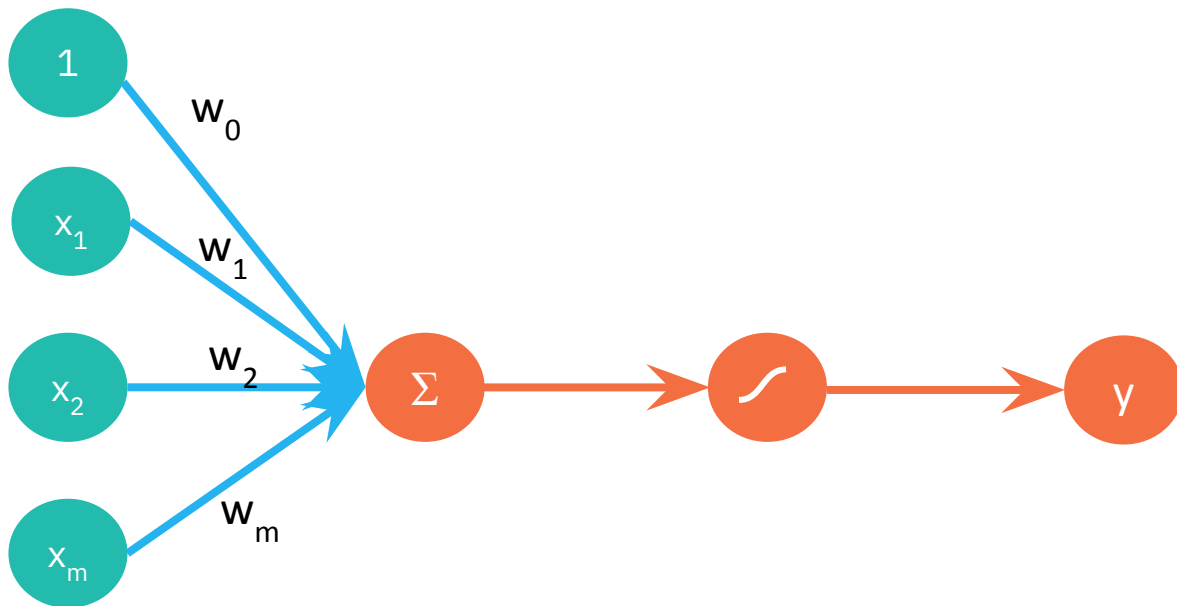
$$y = g(w_0 + X^T W)$$

$$X = \begin{bmatrix} x_1 \\ \dots \\ x_m \end{bmatrix} \quad W = \begin{bmatrix} w \\ \dots \\ w_m \end{bmatrix}$$

Three simple steps:

1. Dot product
2. Add a bias
3. Apply nonlinearity

Inputs Weights Sum Non-linearity Output



Activation Functions

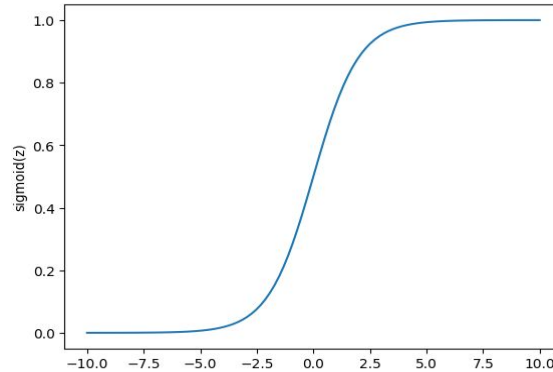
- Sigmoid function

- $g(z) = \frac{1}{1 + e^{-z}}$

- $g'(z) = g(z)(1 - g(z))$

Originally from:

- $\sigma(z) = \frac{L}{1 + e^{-k(z-z_0)}}$



```
import matplotlib.pyplot as plt
import numpy as np
```

```
z = np.linspace(-10, 10, 100)
```

```
def sigmoid(z):
```

```
    return 1.0/(1.0 + np.exp(-z))
```

```
a = sigmoid(z)
```

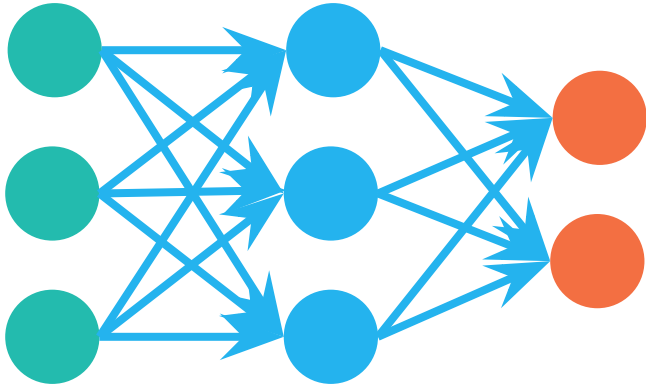
```
plt.plot(z, a)
```

```
plt.xlabel("z")
```

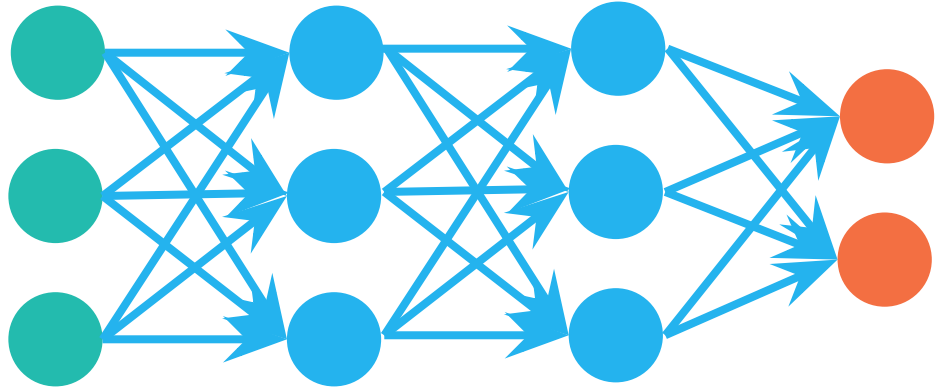
```
plt.ylabel("sigmoid(z)")
```

Building Neural Networks

- Single Layer



- Multi Layer



Homework 1

- Install Visual Studio – VS Code (upgrade your OS first)
- Install Python 3.13 (if you can't install 3.13 then try 3.11)
- For Mac OS: <https://www.youtube.com/watch?v=QmFOI8GYz6E>
- For Windows: https://www.youtube.com/watch?v=cUAK4x_7thA

- For a python tutorial look at: <https://www.codecademy.com/>

Happy Hacking!

Building AI **together**

ZiND!

